Do AI Models Remember Indian Company Financials? Evidence on Look-Ahead Bias by Firm Size

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Keywords: large language models, look-ahead bias, memorization, backtesting, Indian equities, firm size. *JEL codes:* G14, G17, C45, C80.

Abstract

Large language models are increasingly used to forecast earnings and returns, but a model that has memorized past outcomes produces backtests contaminated by look-ahead bias. We test whether three widely used models recall the reported revenue and net profit of 79 Indian listed companies, from ₹123 crore to ₹8.9 lakh crore in market value, for fiscal years 2023 to 2026, scoring answers against first-reported figures. For FY23 and FY24, well before their training cutoff, the Gemini models recall revenue closely and the larger one almost exactly: Gemini 3.6 Flash puts 94% of FY23 and FY24 answers within 5% of the truth, most of them within 1%, and the smaller Gemini 3.5 Flash-Lite 79% and 67%. Recall falls with firm size, and model capacity moves where it falls: Flash-Lite's FY24 accuracy drops from 94% for large caps to 29% for micro-caps, while Gemini 3.6 Flash stays at 96–100% down to small caps and falls to 71% only for micro-caps. Each tenfold increase in market value raises the probability of accurate recall by about 11 percentage points. After the cutoff, accuracy falls to 24–28%, only modestly above a naive random-walk guess (18%). gpt-oss-120b, by contrast, recalls almost nothing (1–6%). LLM-based backtests on Indian equities therefore carry a serious contamination risk that depends on the model and the firm's size.

Use of AI tools

The author ran all model queries and analyses, checked audited figures against the original company filings, spot-checked reported numbers against the outputs, and reviewed the results and the text. Generative AI (Anthropic's Claude) assisted with the study design, wrote the data-collection and analysis code, helped collect and audit the ground-truth figures, and drafted the text. Every reported number was re-checked

against the underlying data in a separate verification pass, and all data, raw model responses and code are public (see Data and code). The author takes full responsibility for the content.

1. Introduction

If an AI model already knows what happened, then a backtest of its predictions cannot be a real test. Researchers and funds now ask large language models (LLMs) to read news, earnings calls and filings and to predict returns or earnings, then evaluate those predictions on historical data. When the evaluation period overlaps the model's training data, the model may simply recall the outcome. This is look-ahead bias, and it makes a strategy look better in the backtest than it can be in live trading.

The problem is documented for US markets. Lopez-Lira, Tang and Zhu (2025) show that GPT-4o reproduces US macroeconomic releases and index levels almost exactly before its training cutoff, that memorization is stronger for prominent stocks, and that recall disappears after the cutoff. We know of no study that measures this for Indian listed companies, or that maps how it changes across the full size range of a market.

India is a useful test case for two reasons. Its listed universe runs from mega-caps covered by dozens of analysts to micro-caps with almost no English-language coverage, so the amount of text a model could have absorbed varies enormously across firms. And Indian equities are a fast-growing target for retail and quantitative strategies built on AI tools.

We ask three questions. Does an LLM recall the reported revenue and net profit of Indian companies? Does recall fall with firm size? And does it vanish for fiscal years the model cannot have seen? We query three models: gpt-oss-120b, an open-weight model with a June 2024 cutoff, and two Gemini models with a March 2026 cutoff, Gemini 3.6 Flash and its smaller sibling Gemini 3.5 Flash-Lite. For each of 79 companies we ask for four fiscal years, FY23 to FY26, so every model faces years before and after its cutoff, and we score the answers against first-reported figures.

The results are stark. For FY23 and FY24, well before their cutoff, the Gemini models recall revenue closely and the larger one almost exactly: Gemini 3.6 Flash puts 94% of FY23 and FY24 answers within 5% of the reported figure, most of them within 1%. Recall falls with firm size, and a larger model pushes near-complete recall further down the size range: for Flash-Lite, FY24 accuracy is 94% for large caps but 29% for micro-caps, while Gemini 3.6 Flash stays above 95% down to small caps. Each tenfold increase in market value raises the probability of accurate recall by about 11 percentage points. After the cutoff, accuracy falls to 24–28%, only modestly above a naive guess that this year's

revenue equals last year's. gpt-oss-120b, by contrast, recalls almost nothing in any year. Gemini's accuracy rises steeply with its stated confidence, mostly because it is confident about the years it remembers.

The paper makes three contributions. First, it gives the first firm-size map of LLM memorization in an emerging market, to our knowledge, and shows that the boundary moves down the size range as model capacity grows. Second, it shows that a model's effective knowledge boundary can differ from its stated cutoff: Gemini behaves as if FY25 results were only partly in its training data, consistent with Google's own caveat. Third, it releases an open benchmark of 79 companies with ground truth partly audited against company filings, and a test any researcher can rerun on a new model before trusting its backtests.

2. Related work

LLMs as forecasters. A fast-growing literature uses LLMs to predict asset prices and fundamentals. Lopez-Lira and Tang (2023) find that ChatGPT's reading of news headlines predicts next-day US stock returns, and Kim, Muhn and Nikolaev (2024) report that an LLM given anonymized financial statements predicts the direction of future earnings about as well as specialized models. Both results raise the question this paper addresses: how much of an LLM's apparent skill on historical data is memory?

Look-ahead bias in LLMs. Glasserman and Lin (2023) compare GPT sentiment strategies on headlines with and without company names. Inside the training period anonymized headlines do better, which they attribute to a "distraction" effect of the model's general knowledge of a firm. Sarkar and Vafa (2024) show that model outputs contain information unknowable at a stated date (for example, COVID-19 appearing in risk factors generated for 2019) and that instructing the model to ignore the future does not remove it; our strict prompt tests the same instruction on financial figures. Lopez-Lira, Tang and Zhu (2025) show that GPT-4o reproduces US macroeconomic releases and index levels almost perfectly before its cutoff, with no recall after it, and that memorization is stronger for prominent stocks. Gao, Jiang and Yan (2025) propose a statistical test of look-ahead bias based on how likely a model finds a prompt, and find that the stronger LLM return predictability among small US stocks is largely look-ahead bias. Benhenda (2026) benchmarks the same problem through trading agents on five large US technology stocks.

Knowledge boundaries. Pęzik et al. (2025) show that models often have several partial knowledge boundaries that differ from their declared cutoffs, so cutoffs must be tested rather than assumed. Our FY25 result is an instance of this in financial data.

What we add. All of these studies use US data. We study an emerging market; we measure recall of hard accounting numbers directly rather than inferring it from trading results; we map recall across the full size range of a market, from ₹123 crore to ₹8.9 lakh

crore; and we score against first-reported rather than restated figures. Our size result complements Gao, Jiang and Yan (2025): direct recall of fundamentals is strongest for large firms, even if return-prediction contamination appears strongest for small ones.

3. Data

The sample is 79 companies listed on the NSE or BSE, drawn from four sectors: IT services, semiconductors and electronics, green energy, and rare earths and metals. We started from 83 companies and dropped four: LTIMindtree (latest annual figures on our source end at FY25), R Systems and ABB India (December year-ends), and Allsec Technologies (stale data).

Ground truth. For each company we record consolidated revenue from operations and consolidated net profit for FY23 to FY26 (years ended 31 March), in ₹ crore, from Screener.in, collected on 19 October 2026 (FY23 on 21 October, checked by matching each company's FY24 figures on the same page). FY23 is excluded for Linde India, whose FY23 covers 15 months, and Owais Metal, which was not yet listed. Screener's net profit includes minority interest. For 12 companies, flagged in the dataset, the figures are standalone rather than consolidated.

Size tiers. We group companies by market capitalisation in October 2026. The thresholds roughly track how AMFI separates large, mid and small caps.

Tier	Market cap (₹ crore)	Companies	Example
Large	1,00,000 and above	17	Tata Consultancy Services
Mid	30,000 to 1,00,000	20	Persistent Systems
Small	5,000 to 30,000	28	Kaynes Technology
Micro	below 5,000	14	Mindteck

Point-in-time audit. A data aggregator shows the latest figures, which may be restated after mergers, demergers or accounting changes. A model learns the figures as first reported. We therefore check 12 companies against each company's own first-reported results (exchange filings and press releases, linked in the dataset): two flagged as suspect (Vedanta, whose FY25 revenue appears restated, possibly because of its demerger, and Coal India, whose FY25 sales figure looks inconsistent) and a random sample of 10 stratified by tier. Of the 72 audited figures (12 companies × 3 years × 2 metrics), 66 match the first-reported number within 1.1%. All six mismatches are later restatements, not errors: discontinued operations (Adani Enterprises FY23 revenue, Coforge FY24 revenue), an accounting change (Coal India FY23 profit), a change in how

levies are presented (Coal India FY25 revenue, 18% higher on Screener), a regrouping (Coal India FY24 revenue) and a restatement around Vedanta's demerger (FY25 revenue, 59% lower on Screener). We replace these six with the first-reported figures, and use exact first-reported decimals for two micro-caps' profits (Sasken and Mindteck), which Screener rounds to whole crores. Two of the ten randomly chosen companies carried a restated figure, so the unaudited part of the sample likely contains some too; this biases measured recall downward, because a model that remembers the original number is scored against a revised one.

4. Method

Models and cutoffs. Indian companies publish annual results in April and May, after the March year-end. We query three models through their free public APIs on 21–22 October 2026: gpt-oss-120b (OpenAI's open-weight model, served by Groq), Gemini 3.6 Flash and Gemini 3.5 Flash-Lite (Google). Their published cutoffs place each fiscal year as follows:

Model	Training cutoff	FY23 (results Apr–May 2023)	FY24 (Apr–May 2024)	FY25 (Apr–May 2025)	FY26 (Apr–May 2026)
gpt-oss-120b (model card)	1 June 2024	before	at the edge	after	after
Gemini 3.6 Flash (model card)	March 2026; some areas only to January 2025	before	before	before (edge in practice)	after
Gemini 3.5 Flash-Lite (model card)	March 2026	before	before	before (edge in practice)	after

FY23 is a year every model could have seen, so it shows each model's best case. FY26 is a year no model could have seen: it is the control, where any accuracy comes from estimation rather than memory. The two Gemini models share a cutoff but differ in size, which isolates the effect of model capacity.

Prompts. Companies are queried in batches of ten per request, because the free tier of the Gemini API allows only 20 requests per day; all models receive identical batches. Each company is named with its exchange code, stating each fiscal year's end date, and fixing the unit (₹ crore). The *strict* prompt asks the model to answer from memory and return null if it does not know. The *best-estimate* prompt asks for a number for every figure, plus a 0–100 confidence. That gives 79 companies × 3 models × 2 prompts = 474 company answers (48 requests), each with 8 figures. The exact prompts are in the public repository (see Data and code). We first planned to use Gemini 3.8 Flash, but its free tier was exhausted by pilot runs and heavily overloaded; Gemini 3.6 Flash has the same training cutoff. We use temperature 0, the lowest reasoning setting each API offers (low for gpt-

oss-120b, minimal thinking for Gemini) and no search tools. We store every raw response with a timestamp; requests that failed because of server overload were retried, and every attempt is kept in the log.

Scoring. For each figure we compute the absolute percentage error:

$$APE = |\hat{y} - y| / |y|$$

where $\hat{y}$ is the model's answer and y the reported figure. The headline measure is the share of answers with APE ⩽ 5% (hit5); we also report APE ⩽ 1% (hit1) and the median APE. A missing answer counts as a miss. Answers off by almost exactly a power of ten, or consistent with US$ millions, are flagged as unit errors and reported separately.

Inference. Confidence intervals for hit rates come from a bootstrap that resamples companies (2,000 draws), since each company contributes several correlated answers. We then estimate a logistic regression of hit5 on log market capitalisation, cutoff status (before, edge, after) and a model indicator, with standard errors clustered by company. A random-walk benchmark (this year's revenue equals last year's) shows how often a naive guess lands within 5%.

5. Results

For FY23 and FY24 the Gemini models recall the revenue of Indian listed companies closely, Gemini 3.6 Flash almost exactly; recall falls sharply with company size; and gpt-oss-120b recalls almost nothing.

5.1 Do the models answer at all? Under the strict prompt, both Gemini models give a revenue figure for every large company in FY23 and FY24, and for none in FY25 or FY26. Gemini 3.5 Flash-Lite becomes more cautious for small firms, answering only 38% of micro-cap FY23 figures and 29% of FY24 figures, against 92–93% for Gemini 3.6 Flash. gpt-oss-120b returns null for every figure in every year, including FY23, which is before its cutoff.

5.2 Accuracy by size and cutoff. Share of revenue answers within 5% of the reported figure, best-estimate prompt (95% bootstrap intervals in brackets):

Model	Tier	FY23	FY24	FY25	FY26
Gemini 3.6 Flash	Large	100% [100–100]	100% [100–100]	71% [47–88]	41% [18–65]
Gemini 3.6 Flash	Mid	95% [84–100]	100% [100–100]	55% [35–75]	15% [0–30]
Gemini 3.6 Flash	Small	100% [100–100]	96% [89–100]	39% [21–57]	14% [4–29]

Model	Tier	FY23	FY24	FY25	FY26
Gemini 3.6 Flash	Micro	69% [46–92]	71% [50–93]	29% [7–50]	36% [14–64]
Gemini 3.6 Flash	All	94% [87–99]	94% [89–99]	48% [37–59]	24% [15–34]
Gemini 3.5 Flash-Lite	Large	94% [82–100]	94% [82–100]	71% [47–88]	35% [12–59]
Gemini 3.5 Flash-Lite	Mid	95% [84–100]	95% [85–100]	45% [25–65]	25% [10–45]
Gemini 3.5 Flash-Lite	Small	71% [54–86]	50% [32–68]	18% [4–32]	29% [14–46]
Gemini 3.5 Flash-Lite	Micro	54% [31–77]	29% [7–57]	14% [0–36]	21% [0–43]
Gemini 3.5 Flash-Lite	All	79% [70–88]	67% [57–77]	35% [25–46]	28% [18–38]
gpt-oss-120b	All	1% [0–4]	3% [0–6]	5% [1–10]	6% [1–11]
Random walk (last year's revenue)	All	—	17%	15%	18%
Random walk (last year's revenue)	Large / Micro	—	35% / 8%	29% / 0%	35% / 14%

Companies per tier: 17 large, 20 mid, 28 small, 14 micro (19 mid and 13 micro in FY23). gpt-oss-120b is below 15% in every tier and year.

Four patterns stand out. First, for FY23–FY24 Gemini's recall is often near-exact, not approximate: for Gemini 3.6 Flash, 88% of large-cap FY23 revenue answers are within 1% of the reported number, and for Flash-Lite 76%. Second, recall falls with size: Flash-Lite's FY24 hit rate drops from 94% for large caps to 29% for micro-caps. Third, model capacity moves the size boundary: Gemini 3.6 Flash recalls 96–100% of small-cap FY23 and FY24 revenue, against 50–71% for Flash-Lite, and still about 70% for micro-caps. Fourth, after the cutoff (FY26) accuracy falls to 24–28%, only modestly above the random-walk benchmark (18% across all firms), which is roughly what extrapolating a known trend would give.

FY25 sits in between (48% and 35%), and under the strict prompt both Gemini models decline to answer any FY25 figure. This matches Google's note that the Gemini 3 family's knowledge in some areas extends only to January 2025, before Indian companies reported FY25 results in April–May 2025. In practice, FY25 behaves as an edge year for Gemini rather than a clean before-cutoff year.

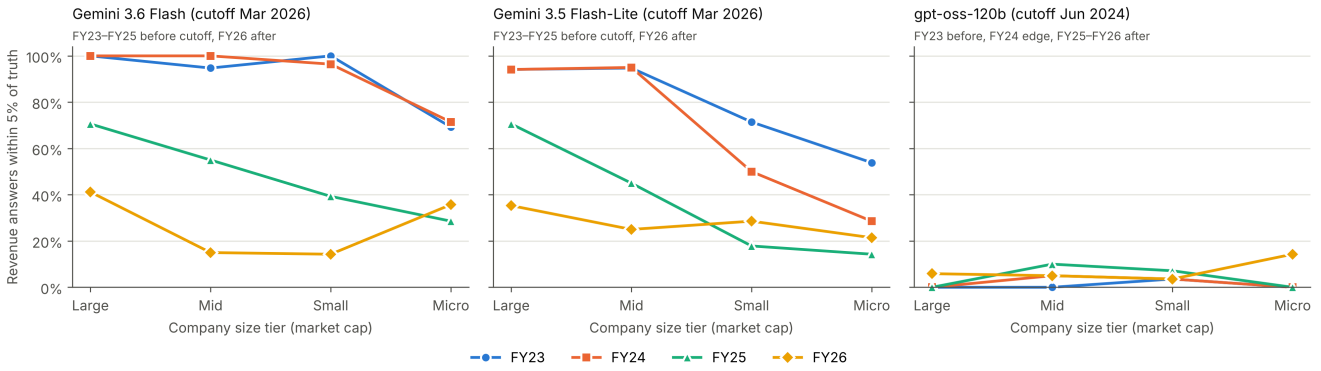


Figure 1. Share of revenue answers within 5% of the reported figure, by size tier and fiscal year, best-estimate prompt. One panel per model; the line under each title gives the cutoff status of each year.

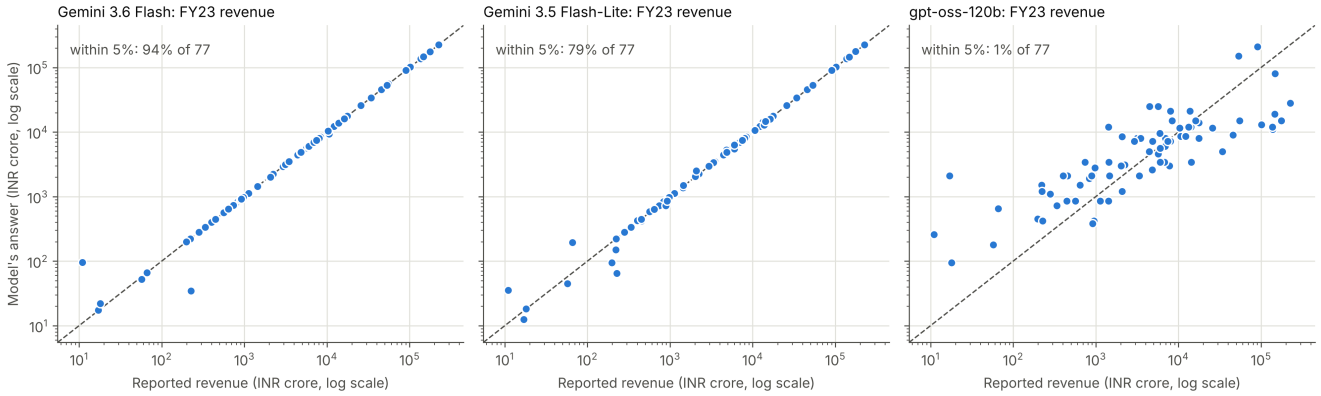


Figure 2. Model answer against reported FY23 revenue, log scales, for the 77 companies with an FY23 figure. FY23 is before every model's cutoff. Points on the dashed diagonal are exact recall.

5.3 Formal test. In a logistic regression of a hit on log market capitalisation, cutoff status and model (942 answers, standard errors clustered by company), a tenfold increase in market value raises the probability of a hit by about 11 percentage points (average marginal effect 0.049 per log unit × 2.30; $p < 0.001$). Being before the cutoff raises it by 25 points relative to after ($p < 0.001$). gpt-oss-120b's FY24 "edge" year is indistinguishable from after-cutoff years ($p = 0.59$). Holding size and cutoff fixed, Gemini 3.6 Flash is 9 points more likely to hit than Flash-Lite ($p < 0.001$), and gpt-oss-120b is 42 points less likely. We code Gemini's FY25 as before its cutoff, as the model card states; coding it as an edge year instead raises the before-cutoff effect to 32 points, gives FY25 a smaller 7-point effect ($p = 0.02$) and leaves the size effect unchanged.

5.4 Does stated confidence mean anything? For the Gemini models, yes. Flash-Lite answers given with confidence 81–100 are within 5% of the truth 86% of the time (113 answers), against 25% at 0–40 (84 answers); for Gemini 3.6 Flash the figures are 99% (135 answers) and 17% (29 answers). gpt-oss-120b's confidence carries no information: it is right 0–10% of the time at every confidence level. Most of the Gemini gradient runs across

years: every Gemini answer given at 81–100 is for FY23 or FY24, and every answer at 0–40 is for FY25 or FY26. Middle levels are overconfident: answers given at 61–80 are right 53–55% of the time.

6. Robustness

The pattern holds under four checks. Net profit shows the same before/after drop at lower levels (Flash-Lite 64% in FY23 to 16% in FY26; Gemini 3.6 Flash 82% to 8%), as expected given its definitional noise. Seventeen answers are off by roughly a power of ten or look like US$ millions (9); 13 are gpt-oss-120b's and 10 are net profit. Rescaling the power-of-ten cases changes no model's overall revenue hit rate in any year by more than 2 points. Dropping the two suspect companies changes no overall hit rate by more than 2 points either. The 1% threshold gives the same ordering, with large-cap FY23 recall of 76–88% for the Gemini models and 0% for gpt-oss-120b.

7. Discussion

For researchers backtesting LLM strategies on Indian stocks. A model that reproduces a large company's revenue to within 1% cannot be asked to "forecast" that company's results for the same year. Any backtest that uses Gemini on Indian large and mid caps for FY23–FY24 is therefore at serious risk of contamination. Small size is no protection with a stronger model: Gemini 3.6 Flash still recalls about 70% of micro-cap FY23 and FY24 revenue figures. Three practical steps follow. Run a recall test like ours on the exact model and sample before backtesting. Restrict evaluation to periods after the model's effective cutoff, which may be earlier than the stated one. Report results separately by firm size.

Model choice matters as much as the cutoff. gpt-oss-120b recalls almost nothing about Indian companies even for FY23, a year before its cutoff, while both Gemini models recall a great deal. A backtest that looks clean with one model may be contaminated with another, so the safety of a test cannot be judged from the cutoff date alone.

Abstention is informative, but incomplete. Under the strict prompt, the Gemini models answer for FY23 and FY24 and decline every FY25 and FY26 figure, yet their best guesses for FY25 are within 5% for 35–48% of companies, well above the random walk. Declining marks clear memories, not all of them. A cheap screen still follows: if a model answers a historical question confidently and exactly, treat its "forecast" for that period as recall; if it declines, do not assume it knows nothing.

8. Limitations

- **Small sample.** 79 companies, and 14 to 28 per tier, give wide confidence intervals. This is a pilot; a full study would cover the NSE 500.
- **Market cap is a proxy.** What matters is how much text about a firm the model saw. Analyst coverage, index membership and listing date are better measures. Several companies listed recently (for example Waaree Energies and Premier Energies in 2024), so little text about them existed before the cutoffs.
- **Three models, one snapshot.** Results may differ for other models and versions; we record the exact version each API served.
- **Batched queries.** Each request asks for ten companies and four years at once, which may encourage smooth, extrapolated series or mix-ups between companies. A follow-up with paid API access should ask one company and one year per request.
- **Ground truth.** Screener figures may not match the first-reported numbers a model learned; our audit covers 12 of 79 companies. For 12 companies we use standalone figures although the prompt asks for consolidated ones, which biases measured recall downward where the two differ.
- **Why gpt-oss-120b recalls so little.** We ran it with low reasoning effort, to test recall rather than estimation, and it may have seen little Indian company data in training. Its near-zero accuracy even for FY23, before its cutoff, means it cannot separate cutoff effects from model effects on its own; the within-Gemini comparison carries that test.
- **Effective cutoff.** Gemini's behaviour in FY25 suggests its effective cutoff for this domain is earlier than the March 2026 date on the model card, as Google's own note about January 2025 implies.
- **Non-random sample.** The 79 companies come from four sectors chosen for an earlier industry research project, not from a random draw of listed firms. The size gradient is measured within this sample; a study of the NSE 500 would test whether it generalises.

9. Conclusion

Anyone using an LLM to backtest a strategy on Indian equities should check, before trusting the results, whether the model can recall the outcomes it is asked to predict. Our test takes an afternoon and a free API key, and the data and code are public.

Data and code

Ground truth, all raw model responses, the scored dataset and both notebooks: https://github.com/sethu55/llm-memorization-india, archived on Zenodo at https://doi.org/10.5281/zenodo.23624226

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Appendix. Sample

The full list of 79 companies, with tier, market capitalisation, all ground-truth figures, audit sources and every raw model response, is in the public dataset (see Data and code).